

No. of Functional Features Included in the Solution

Date	02 October 2026
Team ID	6ab7f31f8c7c66e70672939b
Project Name	EduGenie - Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Question Answering	Answers general knowledge and academic questions with concise, accurate responses using Google Gemini.	qna.py / POST /qa	Done	0.5
2	Concept Explanation	Simplifies complex concepts into easy language using the local LaMini-Flan-T5 model (CPU compatible).	explanation_module.py / POST /explain	Done	0.5
3	Quiz Generation	Generates 3 multiple-choice questions with 4 options and the correct answer from a topic or passage, returned as JSON.	quiz_module.py / POST /quiz	Done	0.75
4	Text Summarization	Condenses long educational passages into short, clear summaries for quick revision.	summary_module.py / POST /summarize	Done	0.5
5	Learning Path Recommendation	Creates a structured beginner-to-advanced plan with topics, timelines and suggested resources for any subject.	learning_path.py / POST /learn/recommendations	Done	0.75
6	Learner Level Selection	Adapts answer depth and language to the chosen level: beginner, intermediate or advanced.	main.py (LEVELS) / index.html	Done	0.5
7	Web Interface	Responsive HTML/CSS page with task selector, text input, submit button and live result display below the input.	templates/index.html , static/style.css	Done	0.75
8	Input Validation, Error Handling & Model Fallback	Rejects empty or over-4000-character input, handles timeouts and malformed AI output, and retries then falls back to a lighter Gemini model when busy.	main.py (handle, clean_json_block)	Done	0.75

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5 (Q&A, Explanation, Quiz, Summary, Learning Path)
Additional / Nice-to-Have Features	3 (Level Selection, Web Interface, Validation & Fallback)
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Responsive web interface with task dropdown; learner level selection
2	Backend / Logic	4	Concept explanation (LaMini-Flan-T5), quiz generation, summarization, learning path
3	Database / Storage	0	Not applicable - the app is stateless and uses no database
4	API / Integration	1	Question answering via Google Gemini API through FastAPI /qa endpoint
5	Security / Authentication	1	Input validation, API key kept in environment variable, error handling and model fallback